# Lab 1 Report: Bayesian Decision Theory

Q1:

① $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

→ since P(x) is the same for all, it can be ignored

$$p(x|w_1)P(w_1) > p(x|w_0)P(w_0)$$

$$\ln p(x|w_1) + \ln P(w_1) > \ln p(x|w_0) + \ln P(w_0)$$

$$g(x) = \ln p(x|w_1) + \ln P(w_1) - [\ln p(x|w_0) + \ln P(w_0)]$$

Substitute gaussian density:

$$p(x|w_j) = \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left(-\frac{(x-\mu_j)^2}{2\sigma_j^2}\right)$$

$$= \ln(\sqrt{2\pi}) - \ln(\sigma_j) - \frac{(x-\mu_j)^2}{2\sigma_j^2}$$

$$g(x) = -\left[-\ln(\sqrt{2\pi}) - \ln(\sigma_0) - \frac{(x-\mu_0)^2}{2\sigma_0^2} + \ln P(w_0)\right]$$

$$gx = -\frac{(x-\mu_1)^2}{2\sigma_1^2} + \frac{(x-\mu_0)^2}{2\sigma_0^2} - \ln\left(\frac{\sigma_1}{\sigma_0}\right) + \ln\left(\frac{P(w_1)}{P(w_0)}\right)$$

Q2:

A Python function was written to design a two-class minimum-error-rate classifier using a single discriminant function. The program first estimates the mean and standard deviation of each class from the training data. It then computes the Gaussian likelihood for each class and multiplies it by the corresponding prior probability. The discriminant function $g(x)$
g(x) is calculated, and the Bayes decision rule is applied by assigning the input feature to the class with the higher posterior probability. This classifier minimizes the probability of classification error when misclassification costs are equal.

Q3:

Another Python program was implemented that takes a feature value xxx as input and returns the posterior probabilities P(w0∣x)P(w_0|x)P(w0∣x) and P(w1∣x)P(w_1|x)P(w1∣x), along with the value of the discriminant function g(x)g(x)g(x). The program computes the Gaussian likelihoods for both classes, calculates the total probability p(x)p(x)p(x), and then applies Bayes' formula to find the posterior probabilities. The discriminant value is also reported to clearly show how the final class decision is made.

Q4:
The optimal threshold μ1\mu_1μ1 was determined for the case where misclassification costs are equal. This threshold occurs at the point where the discriminant function g(x)=0g(x) = 0g(x)=0, meaning both classes have equal posterior probability. The threshold was computed numerically and plotted along with the class-conditional Gaussian distributions. This value represents the best decision boundary for minimizing classification error when both types of mistakes are treated equally.

Q5:
A second threshold μ2\mu_2μ2 was calculated by assigning a higher penalty to misclassifying Class 1 as Class 0. This changes the Bayes decision rule by increasing the risk associated with choosing Class 0. As a result, the decision boundary shifts toward Class 0, making the classifier more likely to choose Class 1. The new threshold was computed and plotted for different penalty values, showing how higher penalties move the decision boundary.

Q6:
Using sepal length as the feature (x=0x = 0x=0), posterior probabilities and discriminant values were computed for the test inputs x1=[3.3,4.4,1.2,5.0,5.7,6.3,1.5]x_1 = [3.3, 4.4, 1.2, 5.0, 5.7, 6.3, 1.5]x1=[3.3,4.4,1.2,5.0,5.7,6.3,1.5]. These values were organized into a table showing P(w0∣x1)P(w_0|x_1)P(w0∣x1), P(w1∣x1)P(w_1|x_1)P(w1∣x1), and g(x1)g(x_1)g(x1). Based on the sign of g(x1)g(x_1)g(x1), each sample was assigned a class label. The results show clear separation for some values but overlap for others.

Q7:
The same procedure was repeated using sepal width as the feature (x=1x = 1x=1). Posterior probabilities, discriminant values, and class labels were computed and tabulated for the same test inputs. Comparing the results with those from sepal length, sepal width showed greater overlap between the two classes. Based on this comparison, sepal length appears to be a better feature for separating Setosa and Versicolour.

Conclusion:
In this lab, Bayesian decision theory was successfully applied to a binary classification problem using Gaussian class-conditional densities. The effect of prior probabilities and misclassification costs on the decision boundary was clearly observed. Introducing higher penalties shifted the threshold and changed classification behavior. Overall, the experiment demonstrated how Bayesian classifiers make optimal decisions based on probability and risk, and how feature choice significantly affects classification performance.